

Coding Standard Documentation – V.1.0

Java Version: **21**

Spring Boot Version: **4.1.0**

Properties: **application.yml**

Naming Conventions:

Element	Casing Style	Example
Classes & Interfaces	PascalCase	UserService, EventController
Methods & Parameters	camelCase	calculateTotalAmount(Long userId)
Constants	UPPER_SNAKE_CASE	MAX_CONNECTION_TIMEOUT
Packages	lowercase.dot	com.portal.userservice.controller
Database Tables	snake_case (plural)	user_accounts, event_registrations
Database Columns	snake_case	first_name, created_at

Core REST Naming Rules:

- a) **Use Nouns, Not Verbs:**
 - GET /api/v1/users, POST /api/v1/events
- b) **Pluralize Collections:**
 - /api/v1/vacancies, /api/v1/certificates
- c) **Hierarchical Sub-resources:**
 - GET /api/v1/events/{eventId}/participations
- d) **API Versioning:**
 - Namespace all endpoints under /api/v{version}/resource.

Standardized Error Response (RFC 7807 Problem Details):

All microservices must return a uniform error payload on failure:

```
{
  "type": "https://portal.domain.com/errors/resource-not-found",
  "title": "Resource Not Found",
  "status": 404,
  "detail": "Event with ID 1024 was not found.",
  "instance": "/api/v1/events/1024",
  "timestamp": "2026-08-04T18:45:00Z"
}
```

Standard HTTP Status Codes Matrix:

Status Code	Category / Name	Description & Usage Standard
200 OK	Success	Standard response for successful GET , PUT , PATCH , or POST actions returning data payloads.
201 Created	Success	Returned when a new entity is created (POST). Should include the created resource payload and optional Location header.
202 Accepted	Success	Request accepted for asynchronous processing
204 No Content	Success	Request processed successfully, but no content is returned (e.g., successful DELETE operation).
400 Bad Request	Client Error	Malformed request syntax, invalid query parameters, or failed structural validation rules.
401 Unauthorized	Client Error	Missing, expired, or cryptographically invalid authentication token (JWT).
403 Forbidden	Client Error	Authenticated user lacks sufficient Role-Based Access Control (RBAC) permissions.
404 Not Found	Client Error	The requested entity ID or URI endpoint route does not exist.
409 Conflict	Client Error	State conflict in business rules (e.g., duplicate email address, double-booking an event seat).
422 Unprocessable	Client Error	Payload is syntactically valid JSON, but violates domain validation rules (e.g., start date is after end date).
429 Too Many Requests	Client Error	Client exceeded API Gateway rate limit or request quota threshold.
500 Internal Error	Server Error	Unexpected runtime exception in microservice logic. Internal trace details masked for security.
502 Bad Gateway	Server Error	Gateway or proxy received an invalid response from downstream microservice.
503 Unavailable	Server Error	Target microservice is down, under maintenance, or circuit breaker is in OPEN state.
504 Timeout	Server Error	Downstream dependency timed out prior to fulfilling internal REST call.

All Sensitive Data Must Be Loaded From An **.env** File.

Example:

```
server:
  port: ${USER_SERVICE_PORT}

spring:
  application:
    name: user-service
  datasource:
    url: jdbc:mysql://${DB_HOST}:${DB_PORT}/alumni_portal_db
    username: ${DB_USERNAME}
    password: ${DB_PASSWORD}
    driver-class-name: org.postgresql.Driver

eureka:
  client:
    service-url:
      defaultZone: ${EUREKA_SERVER_URL}
  instance:
    prefer-ip-address: true
```

Zero Hardcoded Fallbacks: Do not provide default fallback values in **application.yml** for sensitive credentials like JWT secrets or database passwords. Production configurations must fail fast if required **.env** variables are missing.

Recommended Package Structure.

Example:

